

Extracting and Interpreting Browser Artefacts on Windows

A structured analysis of history, stored credentials, cookies, cache and downloads

Windows 11 forensic image | Autopsy 4.23.1 | Five exercises

05

**Browser artefacts
reconstructed into
one activity picture**

HISTORY • CREDENTIALS
COOKIES • CACHE • DOWNLOADS

The project reconstructs browsing activity from five artefact types

OBJECTIVE

Extract, interpret and correlate browser evidence from a Windows forensic environment.

The investigation tests whether browser records can reveal what was visited, saved, cached and downloaded, while preserving clear evidence screenshots for each exercise.

01	History	URLs, timestamps and referrers
02	Credentials	Saved login record
03	Cookies	Domains, names and access times
04	Cache	Locally retained web resources
05	Downloads	File paths and source URLs

The workflow uses one forensic platform with a focused credential check

01 Windows 11

FORENSIC ENVIRONMENT

Source system and browser profile under examination.

02 Autopsy 4.23.1

PRIMARY ANALYSIS PLATFORM

Parsed history, cookies, cache and download artefacts.

03 WebBrowserPassView

CREDENTIAL VALIDATION

Recovered one stored Chromium-based login record.

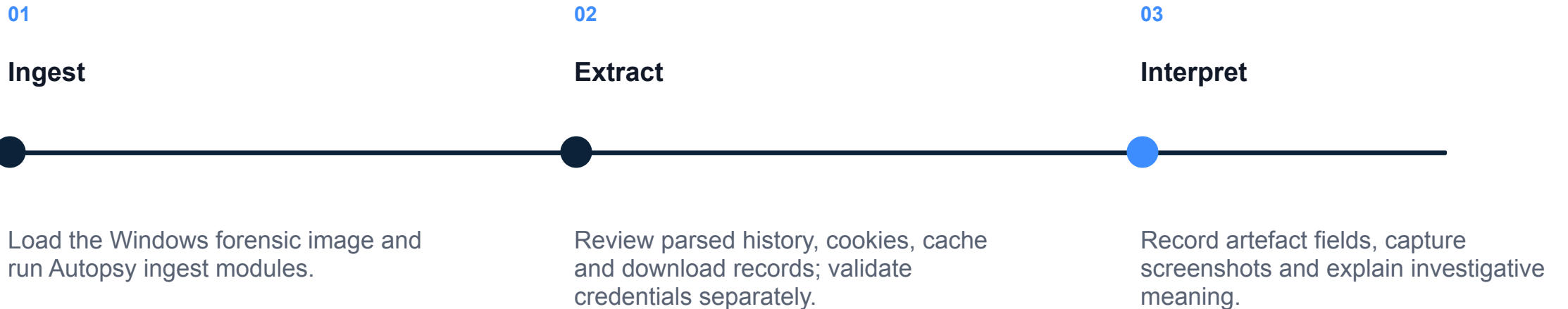
04 DB Browser for SQLite

SUPPORTING TOOL

Available for direct database review when manual validation is needed.

Autopsy replaced separate viewers for most artefacts by parsing the browser databases during ingest.

The method preserves a simple chain from image to interpretation



Forensic principle

Interpret the artefact in context: one record is useful, but correlation across artefact types produces the stronger account of user activity.

History records reconstruct visited sites and their timing

FINDING

73 history records were recovered with URLs, dates and referrer information.

FORENSIC SIGNIFICANCE

The sequence connects Microsoft downloads, Git installation pages, Volatility resources and Bing search activity to specific times.

Visible timestamps include 12 August 2026 at 16:50 and 21:39–21:44 BST.

TableThumbnailSummary

Save Table as CSV

Source Name	S	C	O	URL	Date Accessed	Referrer URL
 History			0	https://www.microsoft.com/en-us/download/details....	2026-08-12 16:50:25 BST	https://www.microsoft.com/en-us/download...
 History			0	https://www.microsoft.com/en-us/download/details....	2026-08-12 16:50:25 BST	https://www.microsoft.com/en-us/download...
 History			0	https://git-scm.com/download/win	2026-08-12 16:50:32 BST	https://git-scm.com/download/win
 History			0	http://git-scm.com/install/windows	2026-08-12 16:50:32 BST	http://git-scm.com/install/windows
 History			0	https://git-scm.com/install/windows	2026-08-12 16:50:32 BST	https://git-scm.com/install/windows
 History				file:///C:/Program%20Files/Git/ReleaseNotes.html	2026-08-12 16:52:39 BST	file:///C:/Program%20Files/Git/ReleaseNote...
 History			0	https://www.volatilityfoundation.org/releases	2026-08-12 21:39:25 BST	https://www.volatilityfoundation.org/release...
 History			0	https://volatilityfoundation.org/releases	2026-08-12 21:39:25 BST	https://volatilityfoundation.org/releases
 History			0	https://volatilityfoundation.org/	2026-08-12 21:44:00 BST	https://volatilityfoundation.org/
 History			0	https://volatilityfoundation.org/about-volatility/	2026-08-12 21:43:43 BST	https://volatilityfoundation.org/about-volati...
 History			0	https://www.bing.com/search?q=SC&pq=volatiity+...	2026-08-12 21:44:00 BST	https://www.bing.com/search?q=SC&pq=v...
 History			0	https://www.bing.com/ck/a?!&&p=da882bbba7123...	2026-08-12 21:44:00 BST	https://www.bing.com/ck/a?!&&p=da882bb...

Evidence: Autopsy Web History screenshot

A stored Chromium login record was recoverable

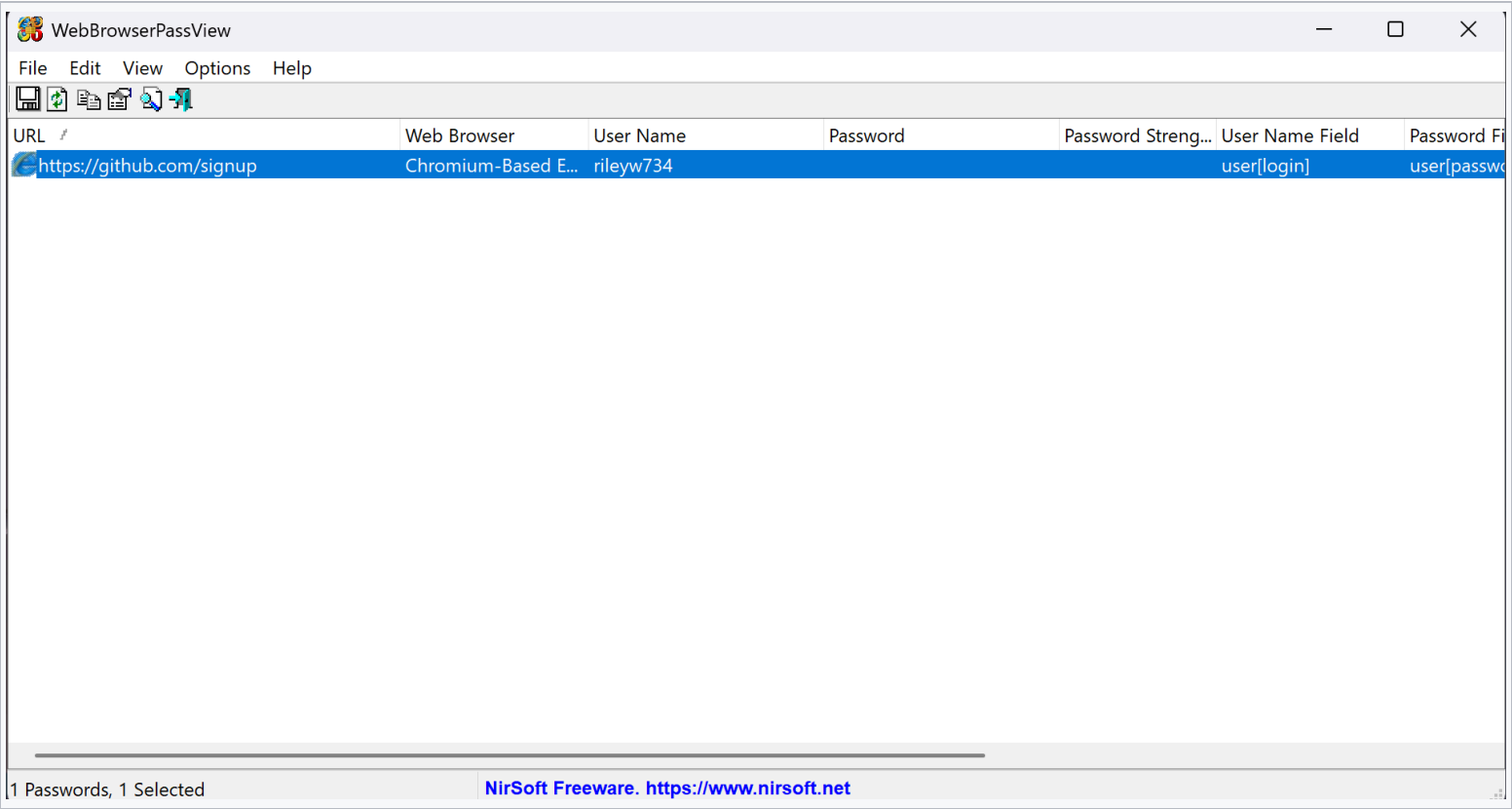
FINDING

WebBrowserPassView identified one saved login associated with the GitHub sign-up page.

FORENSIC SIGNIFICANCE

Stored credential artefacts can reveal account use and create a serious exposure risk when recovery is possible in the correct user context.

The screenshot shows the website and username; no password value is disclosed in the deck.



Evidence: WebBrowserPassView screenshot

Cookies support Microsoft Edge interactions with Bing and MSN

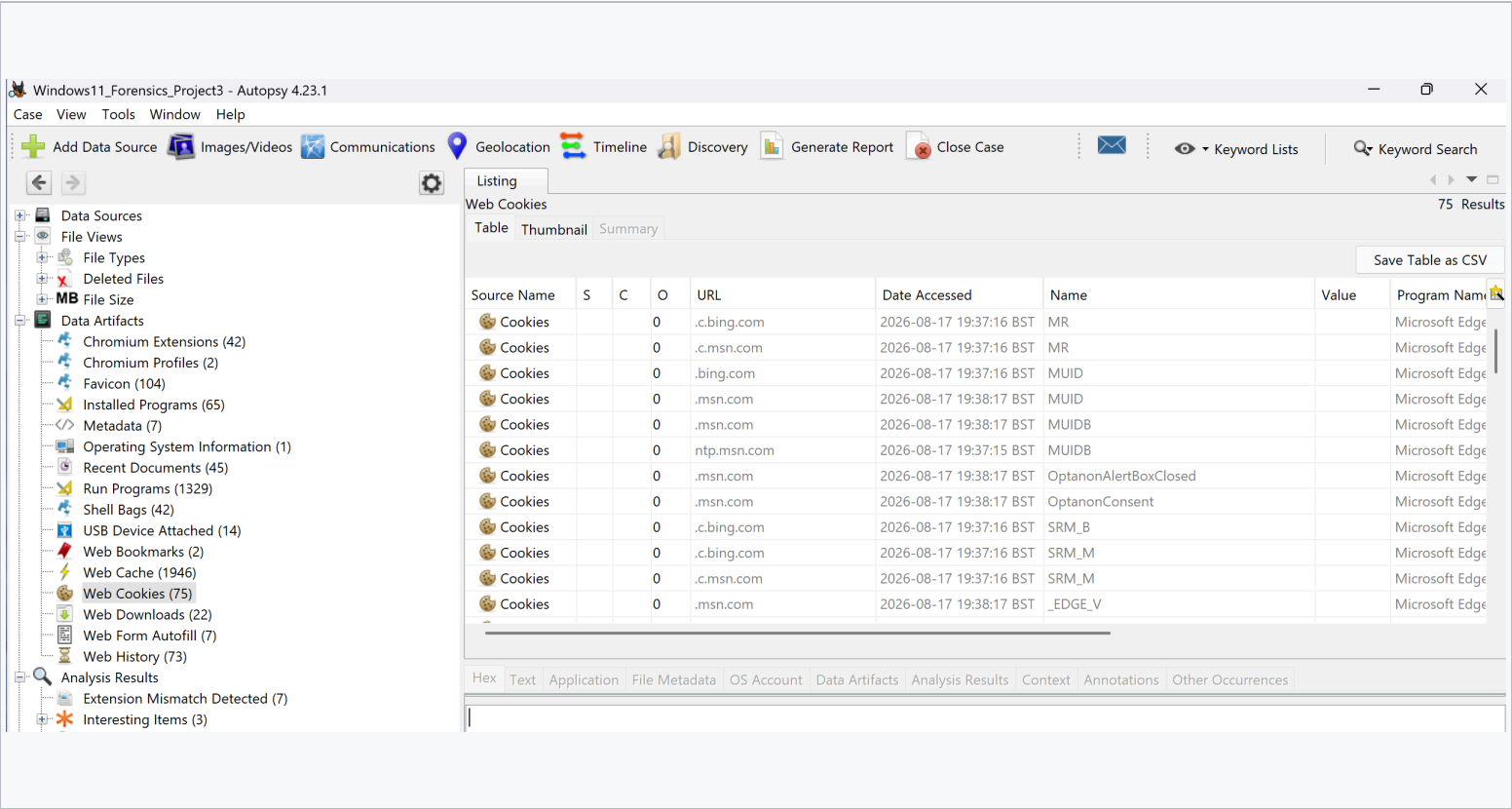
FINDING

75 cookie records were parsed, including Bing and MSN entries from Microsoft Edge.

FORENSIC SIGNIFICANCE

Cookie domains, names and timestamps support evidence of prior web interaction even when full page content is unavailable.

Visible examples include MR, MUID, MUIDB, OptanonConsent and SRM_B.



The screenshot shows the Autopsy 4.23.1 interface with the 'Web Cookies' listing selected. The left sidebar shows a tree view of data sources, with 'Web Cookies (75)' highlighted. The main pane displays a table of 75 cookie records. The table has columns for Source Name, S, C, O, URL, Date Accessed, Name, Value, and Program Name. The records are for various cookies from .bing.com and .msn.com, including MR, MUID, MUIDB, OptanonAlertBoxClosed, OptanonConsent, SRM_B, SRM_M, and _EDGE_V. All cookies are associated with Microsoft Edge.

Source Name	S	C	O	URL	Date Accessed	Name	Value	Program Name
Cookies			0	.c.bing.com	2026-08-17 19:37:16 BST	MR		Microsoft Edge
Cookies			0	.c.msn.com	2026-08-17 19:37:16 BST	MR		Microsoft Edge
Cookies			0	.bing.com	2026-08-17 19:37:16 BST	MUID		Microsoft Edge
Cookies			0	.msn.com	2026-08-17 19:38:17 BST	MUID		Microsoft Edge
Cookies			0	.msn.com	2026-08-17 19:38:17 BST	MUIDB		Microsoft Edge
Cookies			0	ntp.msn.com	2026-08-17 19:37:15 BST	MUIDB		Microsoft Edge
Cookies			0	.msn.com	2026-08-17 19:38:17 BST	OptanonAlertBoxClosed		Microsoft Edge
Cookies			0	.msn.com	2026-08-17 19:38:17 BST	OptanonConsent		Microsoft Edge
Cookies			0	.c.bing.com	2026-08-17 19:37:16 BST	SRM_B		Microsoft Edge
Cookies			0	.c.bing.com	2026-08-17 19:37:16 BST	SRM_M		Microsoft Edge
Cookies			0	.c.msn.com	2026-08-17 19:37:16 BST	SRM_M		Microsoft Edge
Cookies			0	.msn.com	2026-08-17 19:38:17 BST	_EDGE_V		Microsoft Edge

Evidence: Autopsy Web Cookies screenshot

Cache records preserve a broad footprint of web activity

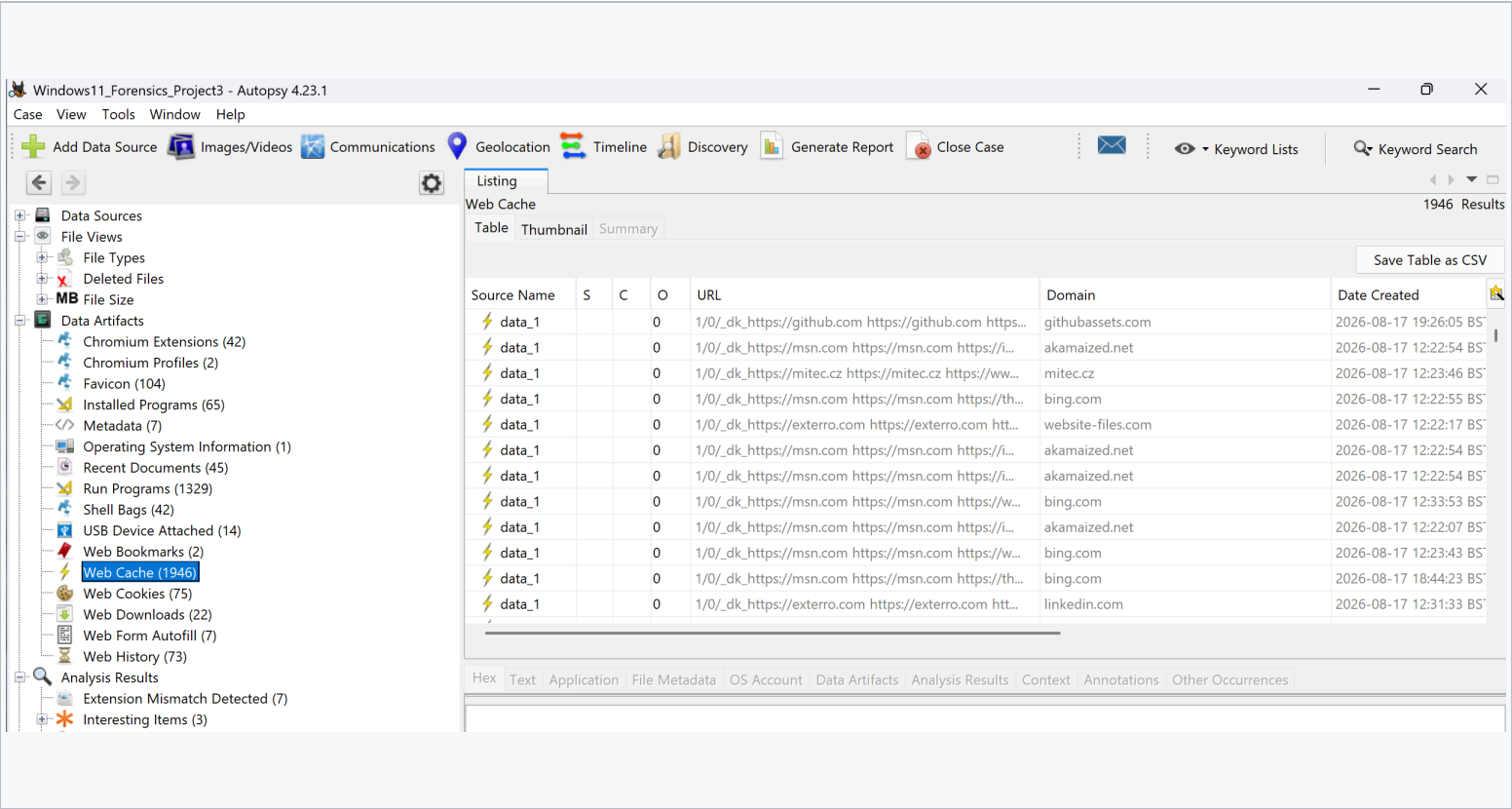
FINDING

1,946 cached entries were recovered with resource URLs, domains and creation dates.

FORENSIC SIGNIFICANCE

Cached resources extend the activity picture beyond history alone and can corroborate access to site assets and third-party content.

Visible domains include githubassets.com, akamaized.net, mitec.cz, bing.com and linkedin.com.



Evidence: Autopsy Web Cache screenshot

Download records link software files to paths and source URLs

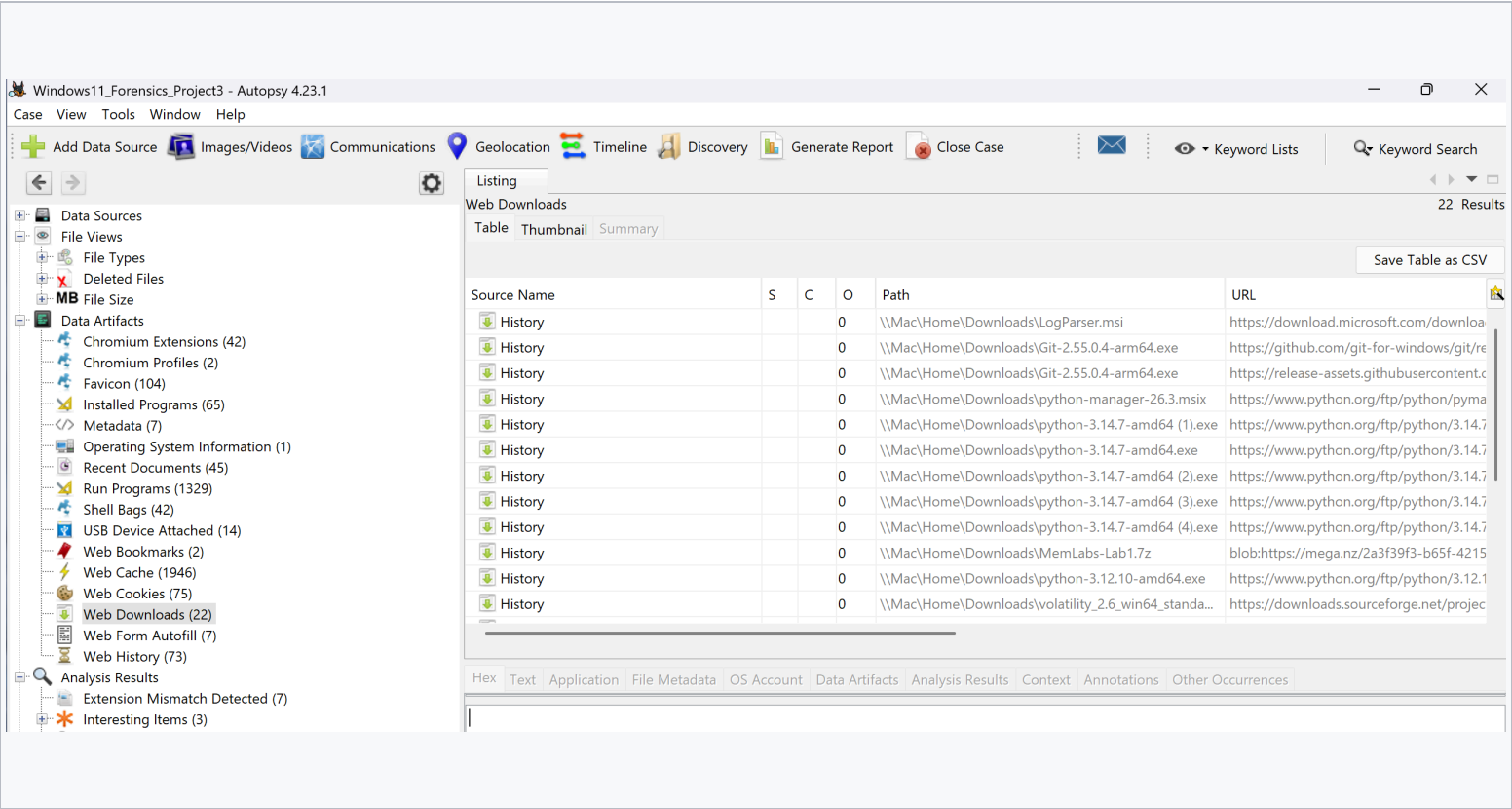
FINDING

22 download records exposed filenames, local paths and originating URLs.

FORENSIC SIGNIFICANCE

The records show which tools entered the environment and where they were stored, supporting both activity reconstruction and follow-up file examination.

Examples include LogParser, Git, Python, MemLabs and Volatility packages.



Evidence: Autopsy Web Downloads screenshot

Five artefact sources tell a consistent activity story

Autopsy recovered extensive browsing evidence, while the credential check added one focused authentication artefact.

73

History records

Visited URLs, timestamps and referrers

75

Cookie records

Edge domains and interaction markers

1,946

Cache entries

The largest recovered evidence set

22

Download records

Files linked to paths and source URLs

1

Stored login

Website and username evidence

Together, the artefacts show browsing, account use, site interaction, locally retained resources and software acquisition.

The evidence meets the outcome of all five exercises

Exercise	Required outcome	Evidence captured	Status
1 — History	URLs with dates and times	Autopsy history table with timestamps and referrers	Met
2 — Passwords	Recover stored browser credentials	One Chromium-based login record	Met
3 — Cookies	Host, name, value and activity fields	Autopsy cookie table with Edge source	Met
4 — Cache	Cached web resources and metadata	Autopsy cache table with domains and dates	Met
5 — Downloads	Files, paths and originating URLs	Autopsy download table with 22 records	Met

Method note: Autopsy parsed the browser databases directly, so separate history/cache/SQLite viewers were not required to achieve the evidential outcomes.

The findings are strong, but artefact availability is never complete

Tool dependency

Results depend on the browser version, user profile, ingest configuration and what remained in the forensic image.

Credential sensitivity

Saved credentials require careful handling. Recovery may fail without the correct user context, and actual passwords should not be displayed.

Interpretation limit

A cache or cookie entry supports prior interaction, but it does not always prove deliberate navigation to a specific page.

Coverage limit

Deleted, cleared, private-mode or synchronised browser data may be missing or incomplete.

Forensic reporting should distinguish the observed record from the conclusion inferred from it.

The strongest conclusions come from correlation, not a single table

01 Preserve context

A URL becomes more useful when paired with a timestamp, referrer, browser source and local path.

02 Correlate artefacts

History, cookies, cache and downloads reinforce one another and reduce reliance on any single record.

03 Choose tools by outcome

A forensic suite can satisfy the brief when it extracts the same evidential fields as separate viewers.

04 Protect sensitive evidence

Credentials and personal identifiers should be minimised, redacted and handled according to the investigation need.

The browser activity was successfully reconstructed across all five exercises.

Autopsy provided a unified view of history, cookies, cache and downloads, while WebBrowserPassView supplied the stored-credential evidence. The combined records demonstrate how browser artefacts reveal user actions, software acquisition and account-related activity.

Final assessment: the project brief is met with traceable screenshots and clear forensic interpretation.